

Garbage Bags and Laundry Baskets

Homemade air flow diagnostic tools get professionally tested.

by **Iain Walker**

Garbage bags can be pretty handy components of an HVAC contractor's tool kit, recent research at Lawrence Berkeley National Laboratory (LBNL) has confirmed (see "Air Flow Measurements in the Bag," *HE* Sept/Oct '02 p. 8). Toting around a laundry basket is also a good idea. Indeed, contractors can often diagnose HVAC system pressures more accurately by using these common household objects than they can using commercially available flow hoods.

Over the last couple of years, the Energy Performance of Buildings Group at LBNL has been investigating methods for improving the measurement of register air flows in homes, after earlier research had found serious precision problems when using commercially available flow hoods. Even the best ones had measurement errors of about 25% and some were much worse. We found that these devices typically had the following problems.

Sensitivity to placement.

Commercial flow hoods need to be centered to work properly. However, their large openings relative to the size of residential registers means that great care must be taken to ensure good centering. Since registers are normally sited near wall/floor, wall/wall, and ceiling/wall intersections, centering flow hoods is virtually impossible in most houses.

Sensitivity to flow non-uniformity. Even when the flow hoods were centered, we found that testing any register that had highly directed flows—thanks to register grille vanes—resulted in relatively inaccurate readings.

Poor resolution at low flows.

With operating flow ranges optimized for commercial grilles, some flow hoods have very poor resolution at the lower residential flow rates. Surprisingly, we found that flow hoods specifically

designed for residential flow rates did not perform significantly better.

Insertion losses. The added flow resistance of the flow hoods reduces the flow through the register being measured. This insertion loss results in an underprediction of flow rates. Most flow hoods offer some sort of correction factor for this effect, but unless the relative flow resistance of different branches is known, it is hard to know what calibration factor to apply. Take the case of a simple two-branch system in which one branch has twice the flow resistance of the other. The correct calibration factor to apply would depend on which branch you were measuring.

Calibration. Some flow hoods had poor calibrations that biased the reading by as much as 50%.

Expense. Finally, commercially available flow hoods typically cost several thousand dollars—a stiff price, especially for an instrument that doesn't necessarily deliver accuracy.

Given these extensive shortcomings, our group looked for alternative measuring devices that would not suffer from these deficiencies. To be practical, any alternative had to be inexpensive and readily available. We came up with a couple that we considered worth pursuing. Garbage bags were our first objects of study. Bag filling is a method of measuring air flow that has been used for some time, but its performance specifications have not been thoroughly determined—an omission we were determined to fix. A second alternative involved an apparatus that improved on a simple flow resistance meter—a step up from a calibrated hole in a box placed over a register.

Bag Filling

We researched the use of garbage bags as a measurement tool for two main reasons. First, bag filling has the advantage that it gives a direct volumetric flow without relying on



Joe Lstiburek and Gary Nelson work with an early air flow measuring device.

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(top) Sylvain Masson measures flow out of a register that is located low on the wall. (bottom) Pulling the cardboard shutter away from the bag causes the bag filling to begin, while the sound of the wood frame contacting the surfaces around the grille gives an audible signal to start the timing.

flow measurement techniques that only sample part of the flow or assume a degree of flow uniformity, as almost all of the other techniques do. Second, as a demonstration of high or low flows to homeowners—or builders and HVAC contractors—bag filling has a direct visual element that is very appealing.

After some experimentation, we implemented two improvements to the method to make the measurements more accurate and more repeatable. One improvement was adding a wood frame to the bag opening. This wood frame ensured that the bag kept its shape and gave a flat surface for easier edge sealing.

The other improvement was adopting a measurement technique that helped to ensure consistent timing. Specifically, the user empties the bag and then places a sheet of cardboard over the bag opening,

between the wood frame and the grille (see photo, above). The cardboard is used as a shutter to make the bag fill as uniformly and rapidly as possible. This assembly is placed close to the grille without blocking the flow. The cardboard is then rapidly pulled away, and the frame of the bag opening is pressed around the grille. The use of the cardboard sheet causes the bag to open rapidly, and the sound of the wood frame contacting the surfaces around the grille gives a consistent audible stimulus to begin timing. We also used a wire frame (made from coat hangers) that we affixed to the bag opening and tested it on a range of entry shapes (again using a cardboard shutter). The bags tend to pop into shape as they become completely filled. This rapid shape change gives a good visual signal to end the timing.

We tested a total of five bags of different sizes and thicknesses. A prime concern was finding the right-sized bags to use. Smaller bags that fill in less than two seconds will produce more errors, because it's hard to get good results in timing how fast the bag is filled. Conversely, bags that are too large for a given register flow will have more leakage around the edges of the bag before it fills completely, and may not generate enough pressure to push the bag into its final shape.

Also, bags made from thinner material often did not fill uniformly, making the experimental results inconsistent. We had success with brand name 30-gallon trash bags and generic garbage bin bags. We had poorer results with heavier bags and a lightweight lawn-and-leaf bag. My best advice is to try out different bags until you find ones that have the right volume and thickness for your application. The following test results refer only to those bags that filled consistently.

Field testing of bags on 30 registers in three houses resulted in biases of -5% and RMS differences of 11% (see “Assessing the Precision of Bags and Baskets”). RMS stands for Root Mean Square and represents a statistical estimate of the difference between the measured and the true values. It is obtained by squaring these differences, averaging them, and then taking the square root of this average.

We found that the largest errors were for grilles whose flows were lower than about 40 CFM. At these small flows, the influence of leakage around the edge of the bag is increased, because it takes a long time to fill the bag, and the small back pressure required to fill the bag becomes more significant at the low-flow grilles. However, it should be noted that the magnitude of the air flow errors in CFM was small for these registers.

Because the bag filling tests require the user to interpret when the bag filling starts and stops, we wanted to look at variability from user to user. Five people (three researchers, a homeowner, and the homeowner's son—none of whom had any previous experience with this measurement technique) measured the same grille three times. Each person was given simple instructions on how to per-

form the test and observed an experienced researcher perform the test. Each person then performed the test three times, and each individual's measurements were then averaged. The five averaged flows fell in a narrow range between 61 and 64 CFM. Compared to the reference flow measurement, there was an RMS error of only 4%.

Laundry Room's Secret Weapon

We also explored alternatives that improved on a simple flow resistance meter. We first experimented with drilling holes in solid containers, but we soon realized that we need look no further than the laundry room for an appropriately vented container—the laundry basket. The baskets work by placing a known flow resistance (the basket) over the register and measuring the pressure difference between the basket and the room. Laundry baskets are about the right size to fit over most registers. They are cheap, have uniform holes, are lightweight, and readily available. We used our laboratory test apparatus to optimize the location of the basket opening and the



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(above) Adding a diffuser screen reduces sensitivity to flow nonuniformity. (right) Holes near the stagnation zone at the base of the basket and close to the opening should be covered with duct tape. For typical residential registers, the optimal area left open should add up to 20 to 40 square inches.



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number of open holes in the basket walls. Normally, a laundry basket has too many open holes to generate a reasonable pressure difference with the air flows out of residential registers. We covered selected holes with duct tape (one use for duct tape that we approve of!). The results were best with the holes left open about halfway up the basket—midway between the stagnation zone at the base of basket and the strongly

directed and nonuniform flow near the entry. We found that laundry baskets that were about 2 ft tall made the air flow more uniformly through the holes. Other improvements included using a soaker hose for pressure sensing (the many small holes give a good average); adding a diffuser screen to further reduce sensitivity to flow nonuniformity; and attaching some weatherstripping around the entry to the basket to improve the seal against the wall (see photos, above).

Assessing the Precision of Bags and Baskets

We first attempted to determine how sensitive the measurements made with garbage bags were to placement over the registers, and to the flow direction out of grilles. The tests used a single duct branch where all the flow goes through the register and flow hood. A fan was used to generate a range of register flows from 50 CFM to 250 CFM. The flows were measured using a high-accuracy ($\pm 0.5\%$) flow nozzle. Four different registers and three placement locations over each register were tested, together with different register damper settings and two different boots (to change the upstream air flow pattern).

For the bags, the errors averaged about 10% of flow averaged over all the configurations. We also tested one bag with a wire frame made from a coat hanger, rather than the wood surround.

We experimented by bending the coat hanger into different opening shapes, to see if the opening shape affected the results. This is important for a couple of reasons. First, users may not be able to create a uniform opening and we need to know how much uncertainty this creates. And second, some grilles in houses are odd shapes or difficult to access (see photo, p. 23), and a flexible frame allows the user to completely cover each grille.

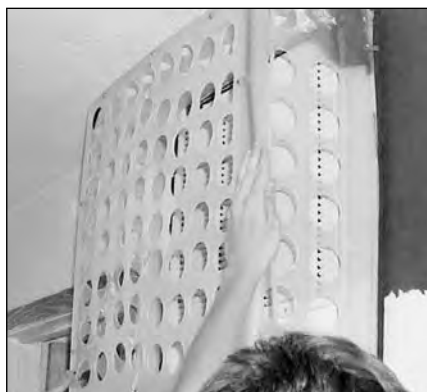
We tested eight opening shapes, using three different grilles: the two-, three-, and four-way-throw grilles. Only the three-way grille showed significant differences—about 5%—for different opening shapes. This was probably because there were more nonuniform flow directions exiting this grille.

The second tests focused on the problem of insertion losses and centering. Two bags were tested on each of the 11 laboratory system grilles three times in order to study both the accuracy of the readings compared to those from the reference flow hood and the repeatability of the results. The RMS errors were about 6% with a -3% bias, with repeatability uncertainties of about 3%.

For the baskets, the accuracy of flow measurement is strongly dependent on the precision of the pressure sensor. We recommend using a digital manometer with 0.1 Pa precision and time-averaging facility. This requires a relatively expensive pressure sensor (about \$500). Because of this requirement, we expect that this flow technique will prove more appropriate for HVAC or weatherization contractors than for homeowners.



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(left) The only way to measure air flow at a register like this is by using a bag with a flexible wire coathanger that can be formed into a shape that fits the available space. (right) Baskets generally measure return air flow quite competently. However, even this large basket did not completely cover this return register, and we had to tape off one row of vents.

Using a series of laundry baskets, we made detailed measurements on 88 registers in three houses and a full-scale laboratory duct system and found that the baskets were pretty accurate flow-measuring devices. They had an average RMS error of about 10% and a bias of 3%. We checked the baskets' results at each grille with an active flow hood that has an accuracy of $\pm 2\%$.

The appropriate number of open holes is a balance of precision and insertion losses. Any flow hood will add a flow resistance to the register and reduce the flow through the register. For greater precision, we want a high pressure signal and therefore fewer holes. On the other hand, to reduce the insertion losses we want more holes. The optimum number of holes for typical residential registers should add up to 20 to 40 square inches. For smaller flows (less than 50 CFM), fewer holes should be opened so as to give a higher pressure signal. For flows greater than 150 CFM, more holes should be opened to reduce the insertion losses.

We tested five different numbers of open holes, ranging from 4 to 48. We determined a general relationship between air flow and pressure difference for each basket by performing a calibration in our duct laboratory. Because most users will not have access to a calibration facility, we determined flow coefficients that could be used together with hole size to determine air flows. The coefficients were slightly different

for each hole opening arrangement, but using the single values given in the following formulas added less than 5% uncertainty to the measurements (this uncertainty is included in the results discussed below). These formulas assume that the measurements are made at sea level with room temperature air.

Air flow (CFM) =
 $\text{Open area} \times 1.252 \times (\Delta P)^{0.5}$;
 for area in in², ΔP in Pa

Air flow (L/s) =
 $\text{Open area} \times 0.09 \times (\Delta P)^{0.5}$;
 for area in cm², ΔP in Pa

To correct for insertion losses, measurements of register flows in eight houses and a laboratory duct system were used to determine an insertion loss correction (ILC) that is based on the measured pressure difference:

$\text{ILC} = (1 + 0.055\Delta P)$
 Corrected air flow = Air flow $\times$ ILC

Incidentally, we found that return registers are even easier to measure with baskets than supply registers, because the returns do not have a strong nonuniform flow. However, return grilles are often much larger than supplies and we were unable to find a suitable basket with holes. Instead, we used a large shallow container and drilled many (128!) holes in it (see photo, above). This number of holes gives a 5

Pa ($1/50$ inch water) pressure drop at a flow rate of 1,000 CFM (472 liters per second). Laboratory tests showed that it has an accuracy of about 1% when used on a single return. The return basket hood was field-tested on five residential and three small commercial buildings. The average RMS difference between the return basket and the reference was 2%. This RMS difference is close to the accuracy of the active flow hood itself, which shows that the return basket hood gives excellent results.

Bags and baskets certainly meet our criteria of being affordable and readily available. They also turn out to be reasonably accurate. Given these characteristics, contractors might want to start thinking of bags and baskets as essential accoutrements.



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For more information:

More details of this testing, plus the testing of commercially available flow hoods, can be found online at <http://epb1.lbl.gov/EPB/Publications/commissioning.html> <http://epb1.lbl.gov/Publications/lbni-47382.pdf>;

<http://epb1.lbl.gov/Publications/lbni-49697.pdf>; and

<http://epb1.lbl.gov/Publications/lbni-51550.pdf>.

For more on garbage bag flow measurement, see the Canada Mortgage and Housing Corporation Web site, www.cmhc-schl.gc.ca/en/burema/gesein/abhos/abhos_ce46.cfm.